

Q1. Explain principle, construction and advantages of a Lithium-ion battery.

- **Principle:** A lithium-ion battery operates on the principle of **intercalation and deintercalation**. During discharge, lithium ions (Li^+) move from the **anode** (negative electrode) through the electrolyte and separator to the **cathode** (positive electrode). At the same time, electrons travel through an external circuit, providing power to the load. During charging, this process is reversed, with lithium ions returning to the anode.
- **Construction:** The primary components are:
 - **Anode:** Typically made of graphite, which acts as a host material for lithium ions.
 - **Cathode:** A lithium metal oxide compound, such as lithium cobalt oxide (LiCoO_2) or lithium iron phosphate (LiFePO_4).
 - **Separator:** A thin, porous membrane that prevents the anode and cathode from touching (which would cause a short circuit) while allowing lithium ions to pass through.
 - **Electrolyte:** An organic liquid containing lithium salts that facilitates the movement of ions between the electrodes.
- **Advantages:**
 1. **High Energy Density:** Stores a large amount of energy in a small, lightweight package.
 2. **Low Self-Discharge Rate:** Holds its charge for a long time when not in use.
 3. **No Memory Effect:** Can be partially charged and discharged without affecting its capacity.
 4. **Long Cycle Life:** Can be charged and discharged thousands of times before significant degradation.

Q2. Define Green Hydrogen Technology. How does it work? Explain with a labeled diagram.

- **Definition:** **Green hydrogen** is hydrogen fuel that is produced without any carbon emissions. It is generated using renewable energy sources, such as solar or wind power, to power the electrolysis process.

- **How it works:** The core of green hydrogen technology is **electrolysis**. An electrolyzer splits water (H₂O) into hydrogen gas (H₂) and oxygen gas (O₂) using an electric current.
 - Anode: Oxygen is produced at the anode.
 - Cathode: Hydrogen is produced at the cathode.
 - Electrolyte: An electrolyte solution or membrane facilitates the transport of ions between the two electrodes.

Since the electricity used is generated from renewable sources, the entire process is carbon-neutral.

- **Diagram:** Imagine a device called an **electrolyzer** connected to a solar panel.
 - The **Solar Panel** (or wind turbine) provides the clean electrical energy.
 - This energy is supplied to the **Electrolyzer**, a container filled with water and an electrolyte.
 - Inside the electrolyzer are two electrodes: an **anode** and a **cathode**, separated by an **electrolyte**.
 - At the cathode, water molecules are reduced to form hydrogen gas (H₂). A pipe or outlet captures this gas.
 - At the anode, water molecules are oxidized to form oxygen gas (O₂). Another pipe or outlet captures this gas.
 - The only output from this process is clean hydrogen and oxygen, with no harmful byproducts.

Q3. Distinguish between Lithium-ion battery and Sodium-ion battery. Write at least 7 points.

Feature	Lithium-ion (Li-ion) Battery	Sodium-ion (Na-ion) Battery
1. Energy Density	Higher. Can store more energy per unit of weight or volume, making them ideal for mobile applications.	Lower. Sodium ions are larger and heavier than lithium ions, leading to lower energy density.

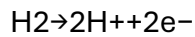
Feature	Lithium-ion (Li-ion) Battery	Sodium-ion (Na-ion) Battery
2. Raw Material	Rare. Lithium is a scarce resource found in limited geographical areas, with high extraction costs.	Abundant. Sodium is highly abundant and can be sourced cheaply from common salt.
3. Cost	Higher. The high demand and scarcity of lithium, cobalt, and nickel make them expensive.	Lower. The abundance of sodium and the use of cheaper materials make them more cost-effective.
4. Anode Material	Typically uses graphite as the anode host material.	Uses cheaper materials like hard carbon or disordered carbon.
5. Operating Temperature	Good performance in a range of temperatures, but can be sensitive to extreme cold.	Exhibits good low-temperature performance and a broader operating temperature range.
6. Thermal Stability	Susceptible to thermal runaway if damaged or overcharged, which can lead to fire.	Generally has better thermal stability and is less prone to catastrophic failure.
7. Voltage	Has a higher average cell voltage (3.6-3.7V).	Has a lower average cell voltage (2.5-3.0V).

Q4. Discuss, in detail, Hydrogen Oxygen fuel cell. Propose two advantages and two disadvantages of fuel cells.

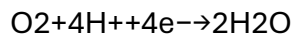
A **Hydrogen-Oxygen fuel cell** is an electrochemical device that converts the chemical energy of hydrogen and oxygen directly into electrical energy. The process is a reverse of electrolysis, with no combustion involved.

- **Working Principle:**

- **Anode:** Hydrogen gas (H₂) is supplied to the anode, where a catalyst splits it into protons (H⁺) and electrons (e⁻).



- **Proton Exchange Membrane (PEM):** The protons pass through a special membrane that acts as an electrolyte. This membrane is permeable to protons but impermeable to electrons and gases.
- **External Circuit:** The electrons cannot pass through the membrane and are forced to travel through an external circuit, generating an electric current that powers a load.
- **Cathode:** Oxygen gas (O₂) is supplied to the cathode. Here, oxygen, protons from the membrane, and electrons from the external circuit combine to form water (H₂O).



The only byproducts are water and heat, making it a very clean energy source.

- **Advantages:**

1. **High Efficiency:** Fuel cells convert chemical energy directly to electricity, making them more efficient than traditional combustion engines.
2. **Zero Emissions:** The primary byproduct of a hydrogen-oxygen fuel cell is pure water, with no harmful greenhouse gases.

- **Disadvantages:**

1. **High Cost:** The catalysts used (typically platinum) are very expensive, making fuel cells costly to manufacture.
2. **Hydrogen Infrastructure:** There is a lack of widespread infrastructure for producing, storing, and distributing hydrogen fuel.

Q5. List various types of solar cells. Compare and contrast Monocrystalline solar panels and Polycrystalline solar panels.

- **Types of Solar Cells:**

- **Crystalline Silicon:** Monocrystalline, Polycrystalline
- **Thin-Film:** Amorphous Silicon (a-Si), Cadmium Telluride (CdTe), Copper Indium Gallium Selenide (CIGS)
- **Emerging:** Perovskite, Organic Photovoltaics (OPV)

Feature	Monocrystalline Solar Panels	Polycrystalline Solar Panels
1. Material	Made from a single, pure silicon crystal.	Made from multiple silicon crystals fused together.
2. Appearance	Have a uniform, dark black color.	Have a fragmented, bluish, or marbled appearance.
3. Efficiency	Higher (17-22%) due to the high purity and uniform crystal structure.	Lower (15-18%) due to impurities at the crystal boundaries.
4. Cost	More expensive to manufacture because of the complex process of growing a single crystal.	Less expensive to manufacture due to a simpler and more cost-effective process.
5. Space Requirement	More space-efficient as they generate more power per square foot.	Less space-efficient and require more panels to generate the same power as monocrystalline.
6. Performance	Perform well in bright, cool conditions.	Perform slightly better than monocrystalline in low-light conditions due to their crystal boundaries.
7. Manufacturing	Uses the Czochralski process to grow a single silicon ingot.	Uses a simpler casting process to create a multi-crystal block.

Q6. How does a fuel cell differ from traditional methods of energy generation? How is a fuel cell different from a battery?

- **Fuel Cell vs. Traditional Energy Generation:**

- **Traditional methods** (e.g., coal, natural gas power plants) rely on **combustion** to generate heat, which is then used to boil water, create steam, and turn a turbine. This is a multi-step process with energy losses at each stage.
- **A fuel cell** directly converts the chemical energy of a fuel (like hydrogen) into electrical energy through an **electrochemical process**, without any combustion. This direct conversion makes it more efficient.
- **Fuel Cell vs. Battery:**
 - **A battery** is an **energy storage device**. It holds a finite amount of energy within its electrodes and must be recharged from an external source when depleted. It is a closed system.
 - **A fuel cell** is an **energy conversion device**. It acts as an open system, continuously converting chemical energy into electricity as long as it is supplied with fuel and an oxidant (like oxygen). It does not "run out" of charge like a battery, but rather runs out of fuel.

Q7. How comparable are Sodium-ion Batteries to Lithium-ion Counterparts?

Interpret with suitable justification.

Sodium-ion batteries are highly comparable to lithium-ion batteries in their fundamental principle, which is the **"rocking-chair" mechanism** of ion movement between electrodes. Both are rechargeable battery technologies.

However, in practice, their performance, cost, and applications make them more of a **complementary alternative** than a direct replacement for Li-ion batteries in all scenarios.

- **Justification:**
 - **Performance:** Sodium-ion batteries have a lower energy density and a lower voltage than Li-ion batteries, which makes them less suitable for high-performance applications like electric vehicles and smartphones, where minimizing weight and volume is critical.
 - **Cost and Sustainability:** Sodium is far more abundant and cheaper than lithium. This makes sodium-ion batteries a highly attractive option for **large-scale, stationary energy storage** applications, such as grid stabilization and renewable energy storage. In these applications, the weight and size of the battery are less important than the overall cost and sustainability.

Q8. Write short notes on the following:

- **(I) Emerging trends of fuel cell:** Trends include the development of more efficient and affordable catalysts to replace expensive platinum, the use of a wider variety of fuels (e.g., natural gas, biomass), and the advancement of high-temperature fuel cells like Solid Oxide Fuel Cells (SOFCs) for combined heat and power generation.
- **(II) Fibre cell:** The term "fibre cell" likely refers to a structural innovation in fuel cell design. This involves using **carbon fibers** as a highly porous and conductive material for electrodes, which can improve the distribution of fuel and the removal of byproducts, leading to increased efficiency and a more compact design.
- **(III) Green chemistry:** Green chemistry is the design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances. Its principles aim to make chemical processes more sustainable, safe, and environmentally friendly from a preventive perspective.
- **(IV) Amorphous silicon solar cell:** This is a type of thin-film photovoltaic cell. Instead of using crystalline silicon wafers, it uses a non-crystalline, disordered form of silicon. Amorphous silicon solar cells are less efficient than crystalline cells, but they are flexible, cheaper to produce, and require less material.

Q9. Elaborate on various applications of the Lithium-ion battery in detail.

- **1. Consumer Electronics:** Li-ion batteries are the dominant power source for portable electronic devices due to their high energy density and compact size. This includes smartphones, laptops, tablets, digital cameras, and power tools.
- **2. Electric Vehicles (EVs):** Li-ion batteries are the heart of the electric vehicle revolution. They provide the power to drive the motor and offer the necessary range for both consumer vehicles and commercial transport. Their light weight is crucial for vehicle performance and efficiency.
- **3. Energy Storage Systems (ESS):** Li-ion batteries are increasingly used for large-scale energy storage. They can store excess energy generated from intermittent renewable sources (like solar and wind) and release it when demand is high, helping to stabilize the power grid.
- **4. Medical and Aerospace Applications:** Their reliability and high energy-to-weight ratio make them ideal for critical applications. They are used in medical

devices like pacemakers and defibrillators, as well as in satellites, drones, and other aerospace technologies.

Q10. Explain the working Principle and advantages of a Sodium-ion battery.

- **Working Principle:** A sodium-ion battery works on the same **rocking-chair** principle as a lithium-ion battery. During discharge, sodium ions (Na^+) move from the anode to the cathode through the electrolyte, and electrons flow through an external circuit. During charging, the process is reversed.
 - **Advantages:**
 1. **Material Abundance and Low Cost:** Sodium is the sixth most abundant element on Earth, found in salt water and salt deposits. This makes it significantly cheaper and more sustainable than lithium.
 2. **Safety:** Sodium-ion batteries are considered to have a better safety profile. They can be stored and transported at zero voltage, which is not possible for Li-ion batteries. They are also less prone to thermal runaway.
 3. **Performance:** They exhibit good performance at low temperatures and are not susceptible to dendrite formation as much as Li-ion batteries. They can be fully discharged without causing damage to the electrodes, which simplifies transportation and recycling.
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Q11. Outline various Principles of Green Chemistry. Illustrate any three in detail.

The **12 Principles of Green Chemistry** are a set of guidelines for chemists to make their processes and products more environmentally friendly.

1. **Prevention**
2. **Atom Economy**
3. **Less Hazardous Chemical Synthesis**
4. **Designing Safer Chemicals**
5. **Safer Solvents and Auxiliaries**
6. **Design for Energy Efficiency**
7. **Use of Renewable Feedstocks**
8. **Reduce Derivatives**
9. **Catalysis**

10. Design for Degradation

11. Real-time analysis for Pollution Prevention

12. Inherently Safer Chemistry for Accident Prevention

- **Illustration of three principles:**

1. **Prevention:** It is better to prevent waste from being created in the first place rather than cleaning it up after it has been created. For example, designing a new chemical synthesis that avoids toxic byproducts eliminates the need for expensive and hazardous waste treatment.
2. **Atom Economy:** This principle advocates for maximizing the incorporation of all materials used in a process into the final product, minimizing waste atoms. A process with 100% atom economy produces no byproducts. Using catalysts is a common way to achieve better atom economy.
3. **Safer Solvents and Auxiliaries:** This principle suggests that chemical processes should avoid the use of hazardous solvents. Instead, safer, non-toxic alternatives like water, ethanol, or supercritical carbon dioxide should be used whenever possible.

Q12. Write short notes on the following:

- **(I) Proton exchange membrane (PEM):** A PEM is a solid polymer electrolyte used in fuel cells, particularly in PEMFCs. It acts as a separator between the anode and cathode, allowing only protons (H^+) to pass through while blocking electrons and reactant gases. This unique property is essential for the fuel cell's operation.
- **(II) Solid oxide fuel cell (SOFC):** An SOFC is a high-temperature fuel cell that uses a solid ceramic material as the electrolyte. It operates at very high temperatures (600-1000°C), which allows it to use a variety of hydrocarbon fuels (like natural gas) and makes it highly efficient.
- **(III) Alkali fuel cell:** This is one of the earliest fuel cell technologies. It uses an aqueous alkaline solution (potassium hydroxide) as the electrolyte. It is highly efficient and operates at relatively low temperatures. However, it is very sensitive to carbon dioxide, which can contaminate the electrolyte.
- **(IV) Drawbacks of fuel cells:** Key drawbacks include their high cost, primarily due to the expensive catalysts (e.g., platinum) and complex manufacturing processes. Other issues are durability, as components can degrade over time,

and the significant challenges associated with storing and transporting hydrogen fuel.

Q13. Classify different types of Electrolyzers in green hydrogen technology. Why green hydrogen is needed?

- **Types of Electrolyzers:**
 1. **Alkaline Electrolyzers (AEL):** Use a liquid alkaline electrolyte (potassium hydroxide) and operate at moderate temperatures. They are a mature and reliable technology, but their dynamic response is slow.
 2. **Proton Exchange Membrane Electrolyzers (PEMEL):** Use a solid polymer electrolyte. They are more compact and have a faster dynamic response, making them a better match for the intermittent nature of renewable energy sources. However, they are more expensive due to the use of platinum-group catalysts.
 3. **Solid Oxide Electrolyzers (SOEL):** Use a solid ceramic electrolyte and operate at very high temperatures. They are the most efficient type of electrolyzer but require a large amount of energy to heat up and are slow to start.
 - **Why Green Hydrogen is Needed:** Green hydrogen is a crucial component of the global effort to decarbonize the energy system. It is a clean, zero-emission fuel that can be used to replace fossil fuels in "hard-to-abate" sectors that are difficult to electrify. These include:
 - **Heavy industry** (e.g., steel and cement production).
 - **Long-haul transportation** (e.g., shipping, aviation, heavy-duty trucks).
 - **Energy storage** for a fully renewable grid.
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Q14. Construct a simple solar cell. Identify different materials used in construction of a solar cell. Analyze In brief the working Principle of a solar cell.

- **Construction of a Simple Solar Cell:** A solar cell is a p-n junction diode designed to absorb light. A simple solar cell consists of:
 - A thin **n-type silicon** layer on top.
 - A thicker **p-type silicon** layer below it.

- A **p-n junction** created at the interface of these two layers, which generates a built-in electric field.
- **Anti-reflection coating** on the top surface to minimize light reflection.
- **Metal contacts** (busbars and fingers) on the top and a full metal contact on the bottom to collect the generated current.
- **Materials Used:**
 - **Silicon:** The most common semiconductor material.
 - **Glass:** To protect the cell.
 - **Metals:** Aluminum, copper, or silver for contacts.
- **Working Principle:** The working principle is based on the **photovoltaic effect**:
 1. **Photon Absorption:** When a photon from sunlight strikes the p-n junction, its energy is transferred to a valence electron, creating an **electron-hole pair**.
 2. **Carrier Separation:** The built-in electric field at the p-n junction separates these generated carriers. The field pushes the electron to the n-type side and the hole to the p-type side.
 3. **Current Generation:** The separated electrons and holes accumulate on opposite sides of the junction, creating a voltage. When an external circuit is connected, the electrons flow from the n-side, through the load, and to the p-side to recombine with the holes, thus creating an electric current.